Progress Report

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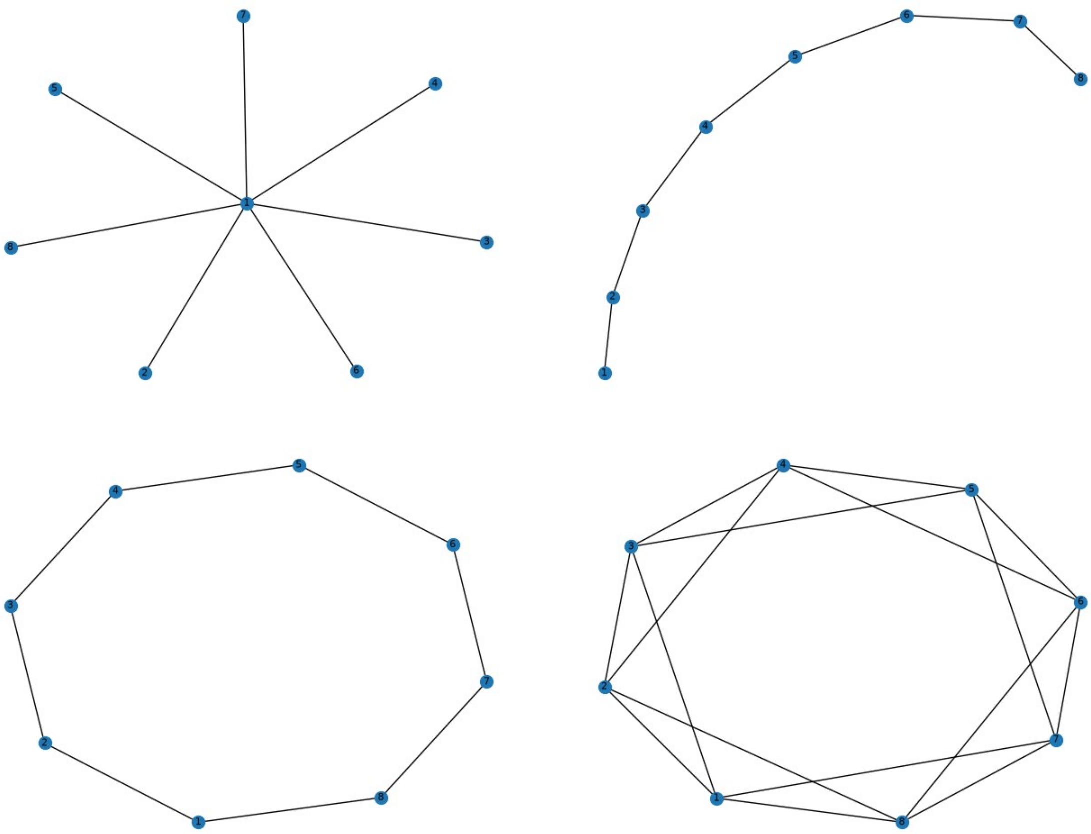
Research Objective and Goal
The final goal of the project is to implement the distributed algorithm for PCA (Principal Component Analysis) developped by our lab and test its validity based on numerical experiment.

Short-Term Goals
10. Continue studying the number of iteration *T_Y* and *T_Z* in the averaging consensus of the matrices *Y_m(t)* and *Z_m(t)* and the number of iteration *T_PM* in the distributed algorithm for PCA alongside with star graphs, path graphs, circle graphs, and regular graphs.

Progress

We will consider 4 different types of networks:
- the star graph
- the path graph
- the circle graph
- the regular graph



I all the previous report, I didn't implemented a stopping condition for the consensus/convergence of the averaging consensus algorithm and decentralized algorithm for PCA. From now, stopping conditions have been set:

- During the averaging consensus algorithm, if every agent’s value is close enough to the convergence value, the algorithm stops.

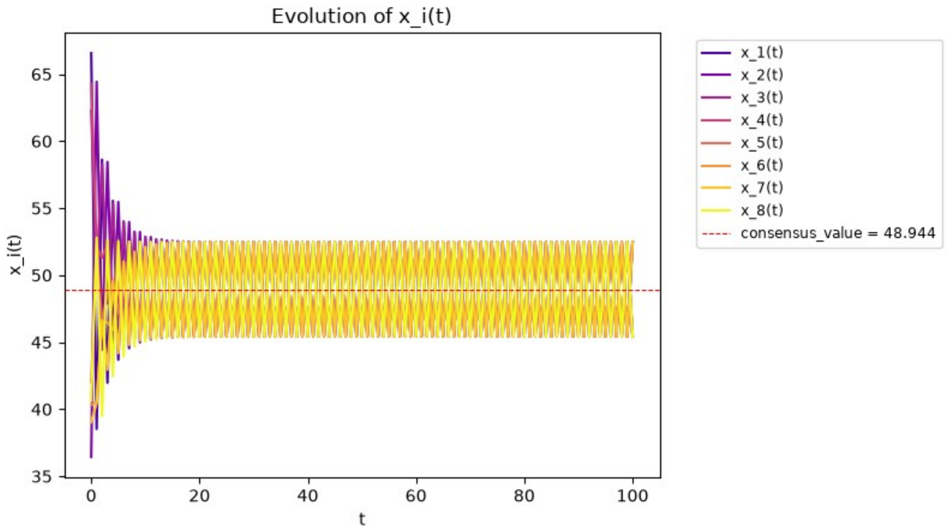
(∀i ∈ {1,…,M}: | xi(t)−xbar(0) | < εconv)
⇒ Convergence,
with εconv > 0

Note that if xi(t) are matrices (and not a scalars), the average matrix xbar(0) is computed element-wise.

- During the decentralized algorithm for PCA, if every agent’s computed matrices is close enough (element-wise) to the computed matrices at the last iteration, the algorithm stops.

(∀m ∈ {1,…,M},∀i ∈ {1,…,N},∀j ∈ {1,…,P}: | U_m(t)[i,j]−U_m(t−1)[i,j] | < εconv)
⇒ Convergence,
with εconv > 0

First of all, after a few preparation tests, I noticed that the first circle graph I generated was unusable for an averaging consensus:



Indeed, the instance of this graph didn’t passed an assertion test for the convergence conditions of the averaging consensus presented in the paper’s algorithm.

2. Convergence conditions

As we have seen, the distributed linear iteration (1) converges to the average, i.e., Eq. (2) holds, for any initial vector $x(0) \in \mathbf{R}^n$, if and only if

$$\lim_{t\to\infty} W^t = \frac{\mathbf{1}\mathbf{1}^T}{n}. \tag{5}$$

We have the following necessary and sufficient conditions for this matrix equation to hold.

Theorem 1. *Eq. (5) holds, if and only if*

$$\mathbf{1}^T W = \mathbf{1}^T, \tag{6}$$

$$W\mathbf{1} = \mathbf{1}, \tag{7}$$

$$\rho(W - \mathbf{1}\mathbf{1}^T/n) < 1, \tag{8}$$

Before proving the theorem, we first give some interpretations:

- Eq. (6) states that $\mathbf{1}$ is a left eigenvector of W associated with the eigenvalue one. This condition implies that $\mathbf{1}^T x(t+1) = \mathbf{1}^T x(t)$ for all t, i.e., the sum (and therefore the average) of the vector of node values is preserved at each step.
- Eq. (7) states that $\mathbf{1}$ is also a right eigenvector of W associated with the eigenvalue one. This condition means that $\mathbf{1}$ (or any vector with constant entries) is a fixed point of the linear iteration (1).
- Together with the first two conditions, condition (8) means that one is a simple eigenvalue of W, and that all other eigenvalues are strictly less than one in magnitude.
- If the elements of W are nonnegative, then (6) and (7) state that W is doubly stochastic, and (8) states that the associated Markov chain is irreducible and aperiodic.

To be more precise, the weight matrix of this network didn’t passed the third convergence condition (8). Note that all generated weight matrices W satisfy the first two convergence conditions (6) (7) by construction. The only remaining convergence condition to be satisfied will always be the third one (8).

Here are the computed eigenvalues in magnitude in descending order of (W-11T/n) and W when using a circle network of 8 agents:

W-11T/n (n=8)	W
1.0000000000000002	**1.0000000000000002**
0.7071067811865483	**0.9999999999999996**
0.707106781186548	0.7071067811865479
0.7071067811865477	0.7071067811865476
0.7071067811865476	0.7071067811865475
1.2454772456427232e-16	0.7071067811865469
5.758675027299075e-17	1.365121385479237e-16
1.032702569292443e-17	2.5324671767710063e-18

We can observe that we have p(W-11T/n) ≥ 1.

We can also note that the third interpretation point says that 1 should be a simple eigenvalue of W. However, according to the table, 1 isn't exactly a eigenvalue of W but 1.0000000000000002 is. A first supposition of this result may be that this is the same problem than the famous result of the addition 0.1+0.2 = 0.30000000000000004, that is to say, this is the consequence of the way floating-point numbers are represented in binary in computers.

So let's suppose that 1 is indeed an eigenvalue of W. But considering this assuption of an eigenvalue being the same as another value if close enough, 0.9999999999999996 may also be in fact the eigenvalue 1. This could means that 1 is an eigenvalue of W of multiplicity 2. And applying this once again on the first column, and now we have that 1 is also an eigenvalue of (W-11T/n), so we have p(W-11T/n) = 1.

For another instance where a circle network of 10 agents has been generated, here are the eigenvalues in magnitude in descending order:

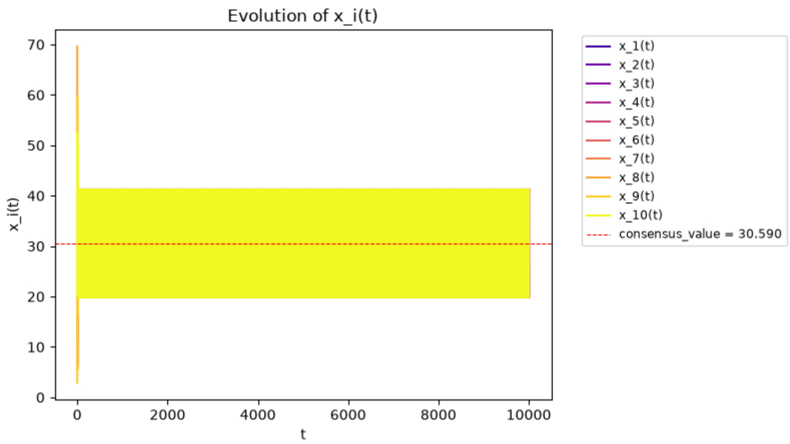
W-11T/n (n=10)	W
0.9999999999999997	**1.000000000000000**4
0.8090169943749482	**1.0**
0.8090169943749479	**0.80901699437494**82
0.8090169943749479	**0.80901699437494**78
0.8090169943749477	**0.80901699437494**77
0.30901699437494773	**0.80901699437494**66
0.30901699437494734	**0.309016994374947**73
0.30901699437494734	**0.309016994374947**5
0.3090169943749473	**0.309016994374947**45
2.028406739448818e-16	**0.309016994374947**34

This instance passed the three assertions of the convergence conditions and we can observe p(W-11T/n) < 1. This means that the three convergence conditions have been verified.

But under the same previous assumption of 0.9999999999999997 and 1.0000000000000004 being in fact the number 1, we have p(W-11T/n) = 1 and 1 being an eigenvalue of W of multiplicity 2 ; Thus, if we assume that p(W-11T/n) = 1, then the third convergence condition is in fact unsatisfied.

We can notice that if we apply the same assumption for some other eigenvalue, we are now left with many eigenvalues of multiplicity greater than one: 0.80901699437494 would be an eigenvalue of multiplicity 4, and 0.309016994374947 would be an eigenvalue of multiplicity 4 in the first column. The same applies for the second column, but also for previous table. We can also notice that the eigenvalues in the two columns are related. However, having eigenvalues of multiplicity greater than one (excluding the greatest eigenvalue) and related eigenvalues between the two matrices might be normal.

Thus, this instance does, technically, indeed, satisfies the 3 convergence conditions in the program, but not in reality as it doesn't converge at all as shown in the plot below.



This is an assumption yet to be confirmed. The may be because of a bug in my program, or (more likely) the consequence of the representation of floating-point numbers in binary, or something else. Thus, the use of a circle graph may be uncertain for now.

However, to raise another point regarding my tests, I noticed that for a circle graph of *NB_AGENT* agents, the averaging consensus seems to converge if and only if *NB_AGENT* is an odd number. In all the tests I conducted, the averaging consensus algorithm had converged when I used a circle network with an odd number of agents, but not when I used a circle network with an even number of agents.

Here's some example for a circle graph of 9 and 11 agents:

W-11T/n (n=9)	W
0.9396926207859083	**1.0000000000000004**
0.9396926207859078	0.9396926207859091
0.7660444431189779	0.9396926207859082
0.766044443118977	0.7660444431189789
0.5000000000000003	0.766044443118978
0.5000000000000001	0.49999999999999994
0.17364817766693044	0.49999999999999933
0.17364817766693022	0.17364817766693036
4.1727013523951153e-17	0.17364817766693022

W-11T/n (n=11)	W
0.9594929736144976	**0.9999999999999999**
0.9594929736144975	0.9594929736144977
0.8412535328311815	0.959492973614496
0.8412535328311812	0.8412535328311814
0.6548607339452859	0.8412535328311805
0.6548607339452845	0.6548607339452848
0.4154150130018866	0.6548607339452844
0.4154150130018861	0.4154150130018868
0.1423148382732852	0.41541501300188644
0.1423148382732852	0.14231483827328512
8.237206509025335e-18	0.14231483827328503

When the number of agents is odd, we can clearly see that p(W-11T/n) < 1 and that 1 is an eigenvalue of W of multiplicity one (under the previous assumption).

So under the assumption that an eigenvalue being in fact the same as another value if close enough (because of the representation of floating-point numbers in binary in computer), we can conclude:
- that circle graphs with an even number of agents do not converge because of the third convergence condition (8) being unsatisfied with p(W-11T/n) = 1.
- that circle graphs with an odd number of agents satisfies the convergence conditions and do converge with p(W-11T/n) < 1.

Circle graphs were a particular case of the study, but it seems that the others type of graph (star/path/regular graphs) shows no particular features, apart from convergence speed (which is normal given the graph topologies).

To talk about the number of iteration needed in circle graphs, I had to set new global variables:
- *CONSENSUS_EPS* for the stopping condition of the averaging consensus algorithm
- *CONVERGENCE_EPS* for the stopping condition of the distributed algorithm for PCA.

I also set initial global variables like previously:
N_DIM = 5, P_DIM = 3
T_PM = 10000, T_Y = 1000, T_Z = 1000
K1 = 0.2, K2 = 0.4, EPS = 0.1
CONVERGENCE_EPS = 1e-10, CONSENSUS_EPS = 1e-8

Let’s change the values of *CONSENSUS_EPS* and *CONVERGENCE_EPS*:

Regular graph (SEED = 15, NB_AGENT = 8, NB_EDGE = 16):

Accuracies (at 1e-i) and average number of iteration needed based on *CONVERGENCE_EPS* (1e-i)

Accuracy CONV_EPS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Iterations	T_Y	T_Z	T_PM
1																49,33	26,00	23,33	1
2																3882,06	25,06	24,08	79
3																18345,64	20,86	24,55	404
4																35354,88	17,63	24,61	837
5																48589,71	15,56	24,63	1209
6																60900,39	13,59	24,64	1593
7																76305,19	11,37	24,64	2119
8																96125,26	9,09	24,65	2849
9																115147,2	7,55	24,65	3576
10																**146241,12**	**5,96**	**24,66**	**4776**
11																281400	3,48	24,66	10000
12																281400	3,48	24,66	10000

Accuracies (at 1e-i) and average number of iteration needed based on *CONSENSUS_EPS* (1e-i)

Accuracy CONS_EPS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Iterations	T_Y	T_Z	T_PM
1																31300	1,07	2,06	10000
2																24334,82	1,19	5,27	3767
3																40841,28	1,46	8,32	4176
4																131900	1,53	11,66	10000
5																165400	1,88	14,66	10000
6																92632,32	3,61	18,32	4224
7																111539,88	4,87	21,35	4254
8																**146241,12**	**5,96**	**24,66**	**4776**
9																158366,94	8,39	28,32	4314
10																180765,88	10,64	31,34	4306
11																205472,96	13,08	34,66	4304
12																232663,21	15,75	38,32	4303

Star graph (SEED = 15, NB_AGENT = 8, NB_EDGE = 7):

Accuracies (at 1e-i) and average number of iteration needed based on *CONVERGENCE_EPS* (1e-i)

Accuracy CONV_EPS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Iterations	T_Y	T_Z	T_PM
1																227	119,00	108,00	1
2																18067,3	116,27	112,43	79
3																86690,32	98,87	115,71	404
4																167801,76	84,30	116,18	837
5																231027,81	74,76	116,33	1209
6																290021,58	65,65	116,41	1593
7																363005,89	54,84	116,47	2119
8																455971,5	43,47	116,52	2850
9																544759,32	35,49	116,55	3583
10																**636821,6**	**29,49**	**116,57**	**4360**
11																730669,7	25,15	116,59	5155
12																824709,12	21,96	116,60	5952

Accuracies (at 1e-i) and average number of iteration needed based on *CONSENSUS_EPS* (1e-i)

Accuracy CONS_EPS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Iterations	T_Y	T_Z	T_PM
1																140100	1,67	12,34	10000
2																284900	1,85	26,64	10000
3																211708,32	3,86	41,61	4656
4																288705,78	6,34	56,60	4587
5																370408,08	9,88	71,60	4546
6																455275,69	14,74	86,59	4493
7																543802,68	21,22	101,59	4428
8																**636821,6**	**29,49**	**116,57**	**4360**
9																735460,12	39,41	131,27	4309
10																844703,04	50,29	145,97	4304
11																958923,55	61,88	160,97	4303
12																1076567,57	74,23	175,96	4303

Path graph (SEED=15, NB_AGENT = 8, NB_EDGE = 7):

Accuracies (at 1e-i) and average number of iteration needed based on *CONVERGENCE_EPS* (1e-i)

Accuracy CONV_EPS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Iterations	T_Y	T_Z	T_PM
1																424,66	226,33	198,33	1
2																33827,8	221,19	207,01	79
3																161547,48	186,35	213,52	404
4																311372,37	157,84	214,17	837
5																427381,5	139,18	214,32	1209
6																534674,52	121,24	214,40	1593
7																667463,81	100,52	214,47	2119
8																837965,92	79,40	214,52	2851
9																1001182,8	65,11	214,55	3580
10																**1174235,2**	**54,75**	**214,57**	**4360**
11																2424700	27,85	214,62	10000
12																2424700	27,85	214,62	10000

Accuracies (at 1e-i) and average number of iteration needed based on *CONSENSUS_EPS* (1e-i)

Accuracy CONS_EPS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Iterations	T_Y	T_Z	T_PM
1																150800	1,35	13,73	10000
2																421100	1,96	40,15	10000
3																745700	5,91	68,66	10000
4																1047900	6,49	98,30	10000
5																1384900	11,49	127,00	10000
6																1704800	14,17	156,31	10000
7																2081400	22,20	185,94	10000
8																**1174235,2**	**54,75**	**214,57**	**4360**
9																1365131,18	73,15	243,88	4306
10																1574360,16	93,19	272,60	4304
11																1796244,32	115,54	301,90	4303
12																2020301,53	138,89	330,62	4303

Circle graph (SEED=15, NB_AGENT = 7, NB_EDGE = 7):

Accuracies (at 1e-i) and average number of iteration needed based on *CONVERGENCE_EPS* (1e-i)

Accuracy CONV_EPS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Iterations	T_Y	T_Z	T_PM
1																326	171,67	154,33	1
2																29203,57	167,30	160,83	89
3																173931,6	135,30	165,10	579
4																672397,36	75,06	165,77	2792
5																1086667,04	58,24	166,00	4846
6																1499310,66	49,04	166,10	6969
7																1512388,32	42.40	166,16	9102
8																2061500	39,98	166,17	10000
9																2061500	39,98	166,17	10000
10																**2061500**	**39,98**	**166,17**	**10000**
11																2061500	39,98	166,17	10000
12																2061500	39,98	166,17	10000

Accuracies (at 1e-i) and average number of iteration needed based on *CONSENSUS_EPS* (1e-i)

Accuracy CONS_EPS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Iterations	T_Y	T_Z	T_PM
1																134100	2,92	10,49	10000
2																354600	2,06	33,40	10000
3																605400	4,96	55,58	10000
4																872100	9,62	77,59	10000
5																1150600	15,30	99,76	10000
6																1439000	22,09	121,81	10000
7																1745700	30,74	143,83	10000
8																**2061500**	**39,98**	**166,17**	**10000**
9																2385500	50,36	188,19	10000
10																2714700	61,28	210,19	10000
11																3049400	72,74	232,20	10000
12																3389500	84,74	254,21	10000

When decreasing *CONVERGENCE_EPS*, we can see that:
- the average required amout of iteration needed to computed matrices Z_m(t) is slightly increasing over the update rule iterations
- the average required amount of iteration needed to compute matrices Y_m(t) is decreasing
- the average required amout of iteration nedeed for the update rule to converge is increasing.

To explain the first point, we first recall that *CONSENSUS_EPS* (the number of iteration affecting the convergence accuracy of the averaging consensus algorithm) is fixed. Thus, there is no big change about the numbers. An explanation regarding why the numbers seems increasing more rapidly at the beginning of the tables would be that the number of iteration of the update rule isn't high enough to compute an accurate average. We can see that it is always an increasing, even for the others graph type. This couldn't be explain because of the fixed *SEED*, since it is also always slightly increasing over the update rule iterations for all the other instances I generated. I don't know why this happens.

To explain the second point, we also recall that *Y(t)* is the average consensus of matrices *U_m(t)*. Since the matrices *U_m(t)* are already converging to the optimal solution, computing *Y_m(t)* from *U_m(t)* would require less and less iterations over the update rule iterations. This would explain why no iteration is in fact necessary to find an approximation of the optimal solution of the problem and why *T_Y* and *T_PM* are related as noticed in the previous report. The *U_m(t)* matrices are naturally converging over the update rule iterations.

To explain the third point, this is simply due to a convergence threshold on the update rule that becomes stricter as *CONVERGENCE_EPS* decrease.

When decreasing *CONSENSUS_EPS*, we can see that:
- the average required amount of iteration needed to compute matrices Y_m(t) and Z_m(t) are increasing.

To explain this point, this is simply due to a convergence threshold on the averaging consensus that becomes stricter as *CONSENSUS_EPS* decreases.

Note that for the star/path graph, the number of iteration needed for the update rule to converge is decreasing as *CONSENSUS_EPS* is decreasing.
Regarding the circle graph, the T_PM column isn't changing. I suppose that the number of iteration needed for the update rule is also increasing, only that the circle graph requies more than 10000 update rule iteration to converge.
Thus, I would assume that the number of iteration needed for the update rule to converge is decreasing as *CONSENSUS_EPS* is decreasing.
However, the registered number of iteration for the update rule regarding the regular graph is changing weirdly. So this may be an incorrect assuption.

To talk about the accuracies, for the regular/star/path graphs, it appears that the found solution has a great accuracy if:
- the *CONVERGENCE_EPS* threshold is less of equal to 1e-10 when *CONSENSUS_EPS* threshold is equal to 1e-8
- the *CONSENSUS_EPS* threshold is less or equal to 1e-8 when *CONVERGENCE_EPS* threshold is equal to 1e-10.

However, regarding the circle graphs type, it seems that it needs a higher number of iteration for the update rule.